Notes: Manconi, Massa, and Yasuda (JFE, 2012)

The Role of Institutional Investors in Propagating the Crisis of 2007–2008

Contents

1. Paper at a Glance

Research question. The paper asks whether institutional investors helped transmit the 2007–2008 crisis from securitized bonds into the corporate bond market. The central mechanism is a portfolio-liquidity channel: once securitized bonds became toxic and illiquid in August 2007, exposed investors - especially open-end bond mutual funds - retained the illiquid securitized bonds and sold more liquid corporate bonds instead.

Main answer. The evidence supports this mechanism. Mutual funds with negative flows and high liquidity needs sold corporate bonds, especially lower-rated corporate bonds, rather than selling securitized bonds. Corporate bonds held before the crisis by investors heavily exposed to securitized bonds experienced larger spread increases and more trading/selling pressure than otherwise similar bonds, including bonds issued by the same firms.

Core data. The paper links several security- and investor-level data sources: Lipper eMAXX institutional bond holdings, CRSP mutual fund data, CDA/Spectrum/13F information, Mergent FISD bond characteristics, and TRACE corporate bond transactions. The key advantage is that investor portfolios can be mapped to individual bonds.

Key empirical object. For each corporate bond, the authors construct *holders' exposure*: the weighted-average share of securitized bonds in the portfolios of mutual funds holding that corporate bond before the crisis. This converts the crisis shock from an aggregate event into cross-sectional variation across corporate bonds.

2. Incremental Contribution

- **Moves crisis propagation beyond banks.** Much crisis research emphasizes bank balance sheets. This paper shows that nonbank institutional investors - bond mutual funds and insurance companies - can propagate shocks across asset classes through portfolio liquidation.
- **Identifies an asset-market contagion channel.** The crisis starts in securitized bonds, but corporate bonds are sold because they are liquid enough to be used as cash substitutes. The transmission is not from common fundamentals alone; it arises from who holds what.
- **Uses bond-level identification with issuer fixed effects.** A major concern is that exposed investors may simply hold riskier firms. The paper compares bonds issued by the same firm but held by differently exposed investors, thereby controlling for issuer-level fundamentals, credit quality, and unobserved firm news.
- **Separates mutual funds from insurance companies.** Mutual funds face investor redemptions and liquidity management problems, while insurance companies face different liabilities and capital regulation. The paper finds the strongest transmission through mutual funds.
- **Explains why low-rated corporate bonds were hit hardest.** Lower-rated bonds were sold more aggressively and suffered larger spread increases when held by investors exposed to

securitized bonds. This links the high-yield corporate spread spike to the securitized-bond shock through institutional portfolios.

3. Mechanism and Hypotheses

The paper's logic has three steps. First, institutional investors bought large amounts of securitized bonds before 2007 because these assets were highly rated and treated as liquid. Second, when securitized bonds became information-sensitive and illiquid, investors did not want to sell them at fire-sale prices. Third, investors needing liquidity sold corporate bonds, particularly bonds that were more liquid or whose sale helped restore portfolio risk/liquidity characteristics.

3.1. Hypotheses

- **H1a: Securitized bonds are not sold first.** Toxic securitized bonds become the asset that investors avoid selling, despite being the original shock.
- **H1b: Lower-rated corporate bonds are sold more.** Junk or lower-rated corporate bonds are more likely to be liquidated than high-grade bonds.
- **H2a: Mutual fund outflows predict sales.** Funds facing more negative contemporaneous flows should sell more corporate bonds.
- **H2b: High-liquidity-need funds sell more.** Funds with high turnover or high flow volatility, used as proxies for liquidity needs, should sell more corporate bonds.
- **H2c: Mutual funds sell more than insurance companies.** Insurance companies have more stable liabilities and therefore less immediate need to liquidate.
- **H3a: Holder exposure predicts yield spread changes and trading volume.** Corporate bonds held by investors more exposed to securitized bonds should suffer larger yield spread increases and more trading pressure.
- **H3b: The effect is stronger for lower-rated corporate bonds.** If funds sell lower-rated corporate bonds disproportionately, the exposure effect should be amplified for junk or low-rated bonds.

4. Identification and Measurement

4.1. Why holder exposure is powerful

The same issuer may have multiple bonds outstanding. If one bond is held by mutual funds heavily exposed to securitized bonds and another bond by less exposed investors, the paper can compare spread changes within the issuer. That removes many alternative stories, such as bad news about the firm, sector, or credit quality that affects all bonds of the same issuer.

4.2. Important variables

- **Holders' exposure:** weighted average securitized-bond portfolio share among mutual funds holding a given corporate bond.
- **InvRating:** an inverse rating measure; larger values correspond to lower credit ratings.
- **DYS:** change in yield spread between the pre-crisis quarter and the crisis window.
- **DTr / LogVol:** measures of trading pressure and trading volume around the crisis onset.
- **Turnover ratio and flow volatility:** proxies for mutual funds' liquidity needs.
- **LowFlow / VeryLowFlow:** contemporaneous fund-flow groups used to detect forced liquidation.

5. Extracted Figures and Tables with Detailed Interpretation

5.1. Figure 1. Sample summary statistics

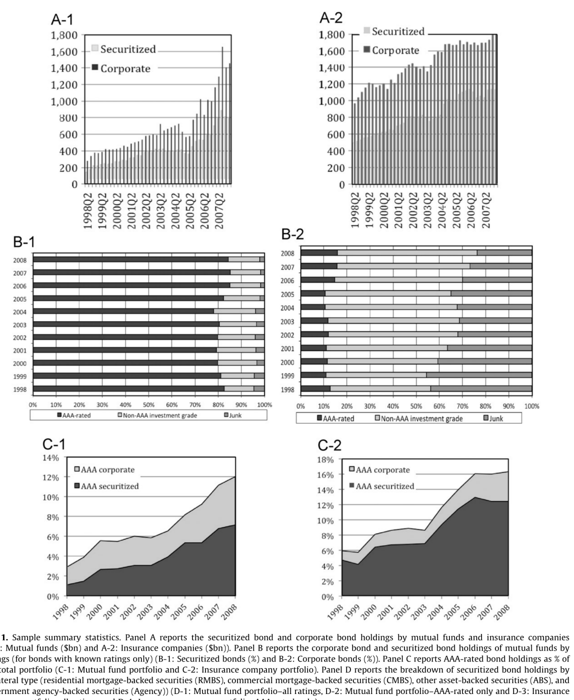

Figure 1, Panels A–C: holdings by investor class, ratings, and AAA share of portfolios.

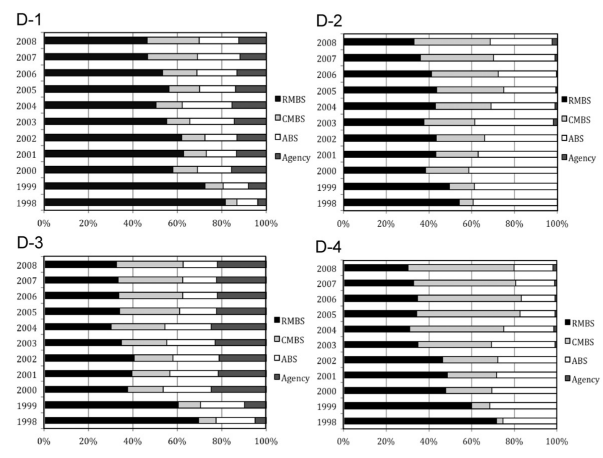

Figure 1, Panel D: collateral-type composition of securitized bond holdings.

Interpretation. Figure 1 establishes the pre-crisis exposure channel. Mutual funds' securitized-bond holdings grew very quickly in the 2004–2007 boom, while insurance-company holdings grew more gradually. The rating panels show that securitized bonds held by institutional investors were overwhelmingly AAA-rated, whereas corporate bond holdings were spread across non-AAA investment grade and junk. This matters because the transmission does not begin with obviously risky low-grade holdings: it starts from assets that were treated as safe and liquid. Panel D also shows that the securitized exposure was not purely agency-backed; private RMBS, CMBS, and ABS were important, making portfolios vulnerable when the whole securitized-bond asset class became suspect.

5.2. Table 1. Sample summary statistics

Table 1
Sample summary statistics.

This table reports the sample summary statistics for the main variables used in the analysis. The variables in Panel A are defined at the fund-quarter level and used in Tables 2–4; the variables in Panel B are defined at the level of individual corporate bonds and used in Tables 2 and 5–7. See Appendix A for variable definitions.

	Mean (1)	Median (2)	St. dev. (3)	Min (4)	Max (5)	No. obs (6)
Panel A: Variables defined at the fund level						
Excess fractional holdings of corporate bonds	0.0335	0.0667	0.2813	−0.8756	0.9862	17,038
Excess fractional holdings of securitized assets	0.0252	−0.0237	0.2498	−0.6290	0.9976	17,038
Turnover ratio (as fraction of TNA)	0.0125	0.0088	0.0103	0.0000	0.0378	16,948
Flow volatility (over 12 monthly flows)	0.0710	0.0326	0.0951	0.0000	0.3733	16,910
Log(Family size) (size in $100M)	3.1136	3.1339	1.8743	0.0000	7.3502	17,038
Affiliated with commercial bank [1=YES]	0.2867	0.0000	0.4522	0.0000	1.0000	17,038
Past flow	0.0160	−0.0002	0.0626	−0.0552	0.2782	16,911
Fund return	0.0133	0.0107	0.0426	−0.3982	0.7702	17,038
Family fractional equity holdings	0.1548	0.0000	0.2718	0.0000	1.0000	17,038
Mgmt fee (%)	0.4923	0.5000	0.2473	0.0000	2.2210	17,038
Expense ratio	0.0105	0.0092	0.0067	0.0000	0.1877	17,038
Actual 12b1	0.0025	0.0000	0.0036	0.0000	0.0103	17,038
Average maturity of holdings (months)	42.7778	37.4805	21.2881	0.0000	196.0000	17,038
No equity [1=NO]	0.8191	1.0000	0.3849	0.0000	1.0000	17,038
Fund's equity holdings return	0.0109	0.0000	0.1029	−0.8961	2.8805	17,038
Panel B: Variables defined at the bond level						
LogSale (July–December 2007) (sale in $K)	4.2004	0.0000	4.7330	0.0000	13.893	9,201
ΔYS(July–October 2007) (%)	0.8151	0.5376	1.2439	−3.998	13.191	7,348
ΔYS(July–December 2007) (%)	1.9650	1.2949	2.0580	−11.310	19.965	8,148
ΔTr(July–October 2007)	0.0375	0.0000	0.0903	0.0000	0.3426	9,133
ΔTr(July–December 2007)	0.0353	0.0000	0.0846	0.0000	0.4997	8,728
Holders' exposure (between 0 and 1)	0.0944	0.0000	0.1472	0.0000	0.9050	9,598
High-turnover holders [1=YES]	0.3113	0.0000	0.4631	0.0000	1.0000	8,728
High-flow volatility holders [1=YES]	0.3291	0.0000	0.4699	0.0000	1.0000	8,728
InvRating	−2.6509	−3.0445	1.0272	−3.3322	0.0000	9,598
Yield spread in 2007Q2 (%)	1.3392	1.1010	1.4089	0.3530	9.4245	9,598
No rating [1=NO]	0.1267	0.0000	0.3327	0.0000	1.0000	9,598
Bond face value (Log($K))	11.275	12.067	2.0613	7.0553	15.425	9,598
Covenants [1=YES]	0.4974	0.0000	0.5000	0.0000	1.0000	9,598
CovIndex (between 0 and 1)	0.1673	0.0000	0.2004	0.0000	0.6667	9,598
Log(Months to maturity)	4.1110	4.2627	1.1467	0.0000	6.9903	9,598
Insurance co.'s exposure (between 0 and 1)	0.2182	0.2761	0.1751	0.0000	0.9716	9,598
Bond is not held by mutual funds [1=NO]	0.4359	0.0000	0.4959	0.0000	1.0000	9,598
Amihud's illiquidity proxy	0.4466	0.4095	0.2848	0.0433	1.5162	9,598
InvTrades	−0.9774	−0.8544	0.5011	−4.0012	−0.2231	9,598

Table 1. Sample summary statistics.

Interpretation. Table 1 shows both the investor-level and bond-level heterogeneity that the paper exploits. On the fund side, there is substantial dispersion in turnover and flow volatility, which allows the authors to compare high- and low-liquidity-need funds. About 29% of sample funds are affiliated with commercial banks and roughly 82% hold no equity, indicating that the main mutual-fund sample is concentrated in bond funds rather than balanced equity-bond portfolios. On the bond side, yield spreads widen sharply during the crisis windows: the average spread increase reaches about 0.82 percentage points by October 2007 and about 1.97 percentage points by December 2007. The mean mutual-fund holder exposure to securitized bonds is about 9.4%, but the maximum is over 90%, which is exactly the kind of cross-sectional exposure variation needed for the paper's tests.

5.3. Table 2. Mutual fund flows and bond sales after the onset of the crisis

Table 2
Mutual fund flows and bond sales after the onset of the crisis.

In Panel A, we examine the relation between contemporaneous fund flows and net position changes. In columns 1–4 the dependent variable is the percentage net purchases of corporate bonds (odd-numbered columns) and securitized bonds (even-numbered). In columns 5–8 the dependent variable is the negative of Log-$ Sales, columns are organized analogously, and the model is estimated with a Tobit regression. The cross section of sample funds are sorted by their contemporaneous fund flows into four bins (*VeryLowFlow*, *LowFlow*, *HighFlow*, and *VeryHighFlow*). In columns 1–2 and 5–6, the net position changes are regressed on just the four category dummies; in columns 3–4 and 7–8, the model also includes additional fund characteristics (see Appendix A for variable definitions). The standard errors are italicized and appear below coefficients. The row labeled *F*-stat reports the *F*-test statistics for the hypothesis that the paired coefficients (corporate bonds vs. securitized bonds) on the most negative flow group (*VeryLowFlow*) are equal. The symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel B reports the estimates of a model:

$$LogSale_{ij} = \alpha + \beta\, BondType_{ij} + \delta\, LogHold_{ij} + \gamma' x_{ij} + \mu_j + \varepsilon_{ij}, \quad (1)$$

where each observation corresponds to a corporate bond. The dependent variable is *LogSale*, the log-net sales (in thousands of dollars) of the bond by institutional investors between July 2007 and December 2007. *LogHold* denotes the log-dollar holding of the bond by institutional investors as of June 2007, μ_j is an issuer-fixed effect for issuer j, and x is a vector of standard bond characteristics, including offering year fixed effects. *BondType* denotes one of three variables of interest: *InvRating* (the natural logarithm of the inverse of 1 + the numerical value of the bond's Standard & Poor's (S&P) rating, which ranges from zero (no rating) to 24 (AAA rating or higher)), the bond's *Amihud* illiquidity ratio, and *InvTrades* (the natural logarithm of the inverse of the average number of daily trades of the bond over the period January–June 2007). In all specifications, the standard errors are clustered around bond issuers and appear italicized. The sample includes all bonds in the FINRA TRACE Corporate Bond Data with available data on bond characteristics from the Mergent FISD Database. The symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Fund flows and bond sales (fund-level analysis)

	Percentage net purchases				Log-$ sales			
	(1) Corporate	(2) Securitized	(3) Corporate	(4) Securitized	(5) Corporate	(6) Securitized	(7) Corporate	(8) Securitized
VeryLowFlow	−0.0461**	−0.0126	−0.1039***	−0.0017	−5.5307***	−0.5190	−5.0510***	0.3837
	−2.30	*−0.40*	*−3.10*	*−0.04*	*−3.49*	*−0.36*	*−3.17*	*0.28*
LowFlow	0.0061	0.0278	−0.0559*	0.0342	−3.5071**	0.9564	−3.1398*	1.5937
	0.40	*1.32*	*−1.81*	*1.08*	*−2.14*	*0.66*	*−1.92*	*1.16*
HighFlow	0.0584***	0.0339	−0.0033	0.0395	−1.4884	2.6290*	−1.3548	3.5581**
	3.22	*1.18*	*−0.11*	*1.08*	*−0.87*	*1.71*	*−0.80*	*2.45*
VeryHighFlow	0.1149***	0.0740***	0.0443	0.0732**	2.0973	3.6415**	2.2739	4.0179***
	5.68	*2.70*	*1.36*	*2.17*	*1.22*	*2.46*	*1.33*	*2.90*
Secur. holdings 2007Q2				−0.1326**				−10.7325***
				−2.17				*−6.14*
Corp. holdings 2007Q2			0.0601*				−4.7101***	
			1.81				*−2.64*	
Affil. comm. bank			−0.0640***	−0.0420*			−1.8665*	−0.0802
			−3.47	*−1.67*			*−1.90*	*−0.08*
Log(Family size)			0.0022	0.0101*			0.0872	0.1491
			0.65	*1.76*			*0.46*	*0.84*
Fam. equity hold.			−0.1076***	−0.0077			−5.7572***	−0.5048
			−3.68	*−0.19*			*−3.71*	*−0.28*
Av. maturity of holdings			0.0009***	0.0004			−0.0047	0.0702***
			2.76	*0.60*			*−0.20*	*3.35*
F-stat (*p*-value)	1.06 (0.3035)		5.06** (0.0245)		6.58** (0.0103)		8.45*** (0.0037)	
St. error	White	White	White	White	White	White	White	White
No. obs.	546	546	546	546	546	546	546	546
(Pseudo-)R^2	0.09	0.02	0.14	0.04	0.02	0.01	0.03	0.02

Panel B: Determinants of corporate bond sales (bond-level analysis)

	(1)	(2)	(3)	(4)
InvRating	1.0496**			1.0712**
	2.14			*2.25*
Amihud		−0.9552***		−0.7488***
		−4.54		*−3.56*
InvTrades			−0.4114***	−0.2938***
			−5.26	*−3.72*
No rating [1=NO]	−3.1242**	0.0897	0.0967	−3.2439**
	−2.01	*0.95*	*1.02*	*−2.15*
Bond is not held by institutional investors [1=NO]	4.8895***	4.6887***	4.7389***	4.6379***
	25.10	*24.01*	*25.04*	*24.34*
LogHold (2007Q2)	1.0148***	0.9738***	0.9840***	0.9647***
	33.00	*31.90*	*32.83*	*31.98*
Bond face value (orthogonalized)	0.1919***	0.1640***	0.1756***	0.1492***
	5.99	*4.76*	*5.27*	*4.29*
Covenants [1=YES]	1.1209***	1.0253***	1.0170***	1.0459***

Table 2, first part: flow groups, net purchases, and sales.

Table 2 (*continued*)

Panel B: Determinants of corporate bond sales (bond-level analysis)

	(1)	(2)	(3)	(4)
	4.61	*4.33*	*4.37*	*4.36*
CovIndex	−0.1055	0.0722	0.0346	−0.1165
	−0.25	*0.20*	*0.09*	*−0.27*
Log(Months to maturity)	−0.3952***	−0.3226***	−0.3886***	−0.3348***
	−9.23	*−8.39*	*−9.58*	*−8.78*
Issuer fixed effects	Yes	Yes	Yes	Yes
Offering year fixed effects	Yes	Yes	Yes	Yes
Standard error cluster	Issuer	Issuer	Issuer	Issuer
No. obs.	9,201	9,078	9,113	9,078
R^2	0.91	0.88	0.91	0.91

Table 2, continued: determinants of corporate bond sales.

Interpretation. Panel A directly supports the liquidity mechanism. Funds in the most negative flow group sell corporate bonds aggressively, while sales of securitized bonds are much weaker. This is exactly the behavior predicted when securitized bonds become too illiquid to sell without large losses: funds meet liquidity needs by selling relatively more liquid corporate bonds. The F-tests comparing corporate-bond sales with securitized-bond sales for the worst-flow funds reject equality in several specifications, which means the liquidation was not proportional across asset classes. Panel B shows that corporate bond sales are concentrated in lower-rated bonds and in bonds that are easier to trade. The message is that funds did not simply sell at random; they selected corporate bonds that could raise cash or reduce portfolio risk when securitized bonds were impaired.

5.4. Table 3. Funds' liquidity needs and pre-crisis holdings

Table 3
Funds' liquidity needs and pre-crisis holdings.
The table reports the estimates of a model:

$$H_{it} = \alpha + \beta \text{ Flow volatility(or turnover)}_{it} + \gamma' x_{it} + \varepsilon_{it}, \tag{2}$$

where each observation represents the portfolio composition of a given mutual fund in a given quarter. The dependent variable H is the excess percentage ownership of the fund's portfolio represented by securitized bonds (Panel A) or corporate bonds (Panel B). *Turnover ratio* and *Flow volatility* are as defined in Section 3.2. x is a set of standard control variables (see Appendix A for variable definitions). In both panels, in columns 1–2 the model is estimated using the Fama-MacBeth procedure. The standard errors are italicized, appear below coefficients, and are Newey-West, with lag length parameter equal to four. In columns 3–4, the model is estimated as a pooled OLS with quarter fixed effects, and standard errors (also shown italicized) are clustered around each fund. The sample includes all the mutual funds belonging to the merged Lipper eMAXX-CDA/Spectrum data set, over the period 1998Q1–2007Q2. The symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

Panel A: Holdings of securitized bonds

	Fama-MacBeth		Pooled OLS	
	(1)	(2)	(3)	(4)
Turnover ratio	2.7576***		2.8124***	
	6.26		*4.94*	
Flow volatility		0.1125**		0.0676
		2.44		*1.12*
Log(family size)	0.0060**	0.0088***	0.0055	0.0079**
	2.44	*3.37*	*1.5*	*2.15*
Affiliated with commercial bank	0.0269***	0.0286***	0.0257*	0.0275*
	2.92	*3.19*	*1.75*	*1.87*
Past flow	−0.0803	−0.1966**	−0.1050*	−0.1655**
	−1.26	*−2.64*	*−1.74*	*−2.07*
Fund return	−0.4052	−0.3836	−0.1813***	−0.1919***
	−1.6	*−1.49*	*−3.68*	*−3.84*
Family equity holdings	−0.0339***	−0.0378***	−0.0376**	−0.0440***
	−4.55	*−5.06*	*−2.35*	*−2.75*
Mgmt fee	−0.0507***	−0.0572***	−0.0774***	−0.0831***
	−5.13	*−5.9*	*−3.07*	*−3.34*
Expense ratio	−4.2264***	−3.9102***	−3.5381**	−3.1500**
	−3.86	*−3.76*	*−2.39*	*−2.19*
Actual 12b1	−1.3601	−1.8687	−2.0375	−2.7213
	−0.85	*−1.28*	*−0.89*	*−1.21*
Av. maturity of holdings	0.0002	0.0004	0.0008**	0.0010***
	0.5	*0.84*	*2.06*	*2.71*
No equity	0.1232***	0.1336***	0.1385***	0.1500***
	10.81	*11.51*	*10.75*	*11.71*
Fund's equity hold. return	0.0224	0.0257	−0.0036	−0.0029
	1.1	*1.18*	*−0.31*	*−0.24*
Quarter fixed effects	No	No	Yes	Yes
Standard error	Newey-West	Newey-West	Clustered by fund	Clustered
by fund				
No. obs.	16,294	16,293	16,294	16,293
(Average) R^2	0.17	0.16	0.14	0.13

Panel B: Holdings of corporate bonds

	Fama-MacBeth		Pooled OLS	
	(1)	(2)	(3)	(4)
Turnover ratio	−2.9380***		−3.1865***	
	−5.66		*−4.94*	
Flow volatility		−0.0599		−0.0018
		−1.04		*−0.03*
Log(family size)	0.0006	−0.0023	0.0017	−0.001
	0.25	*−0.87*	*0.41*	*−0.22*
Affiliated with commercial bank	−0.0267***	−0.0295***	−0.0243	−0.0271
	−2.90	*−3.25*	*−1.45*	*−1.62*
Past flow	0.1538**	0.2125**	0.1564**	0.1459
	2.61	*2.29*	*2.22*	*1.56*
Fund return	0.331	0.3041	0.1666**	0.1797***
	1.04	*0.94*	*2.55*	*2.73*
Family equity holdings	−0.0393***	−0.0341***	−0.0356	−0.0274
	−3.39	*−2.83*	*−1.49*	*−1.15*
Mgmt fee	0.0276**	0.0352**	0.0559**	0.0628**
	2.22	*2.60*	*2.00*	*2.28*
Expense ratio	6.1217***	5.7437***	5.1174***	4.6354***

Table 3, first part: holdings of securitized bonds and beginning of corporate-bond panel.

Table 3 (*continued*)

Panel B: Holdings of corporate bonds

	Fama-MacBeth		Pooled OLS	
	(1)	(2)	(3)	(4)
	4.64	*4.62*	*3.43*	*3.20*
Actual 12b1	−0.1937	0.3661	0.8121	1.5681
	−0.10	*0.21*	*0.32*	*0.63*
Av. maturity of holdings	−0.0010*	−0.0012**	−0.0018***	−0.0020***
	−1.89	*−2.11*	*−4.36*	*−5.00*
No equity	0.1028***	0.0913***	0.0603***	0.0466**
	3.45	*2.99*	*2.90*	*2.24*
Fund's equity hold. return	−0.0143	−0.0186	0.0402**	0.0396*
	−0.21	*−0.27*	*1.98*	*1.92*
Quarter fixed effects	No	No	Yes	Yes
Standard error	Newey-West	Newey-West	Clustered by fund	Clustered by fund
No. obs.	16,294	16,293	16,294	16,293
(Average) R^2	0.10	0.09	0.06	0.05

Table 3, continued: holdings of corporate bonds.

Interpretation. Table 3 answers an important precondition question: were funds with higher liquidity needs more exposed to securitized bonds before the crisis? Yes. The turnover ratio is strongly positively related to holdings of securitized bonds in both Fama-MacBeth and pooled OLS specifications. Flow volatility is also positive in the Fama-MacBeth specification. This means funds catering to investors with more liquid redemption behavior held more of the asset class that turned illiquid in 2007. The table also shows that bank-affiliated funds tended to hold more securitized bonds. Overall, the table supports the sorting interpretation: before the crisis, high-liquidity-need mutual funds bought securitized bonds because they appeared to be liquid, highly rated fixed-income instruments.

5.5. Table 4. Changes in mutual fund holdings of securitized and corporate bonds at the onset of the crisis

Table 4

Changes in mutual fund holdings of securitized and corporate bonds at the onset of the crisis.

Panel A reports the estimates of a model:

$$\Delta B_i = \alpha + \beta \text{Flow volatility(orturnover)}_i + \gamma' x_i + \varepsilon_i. \quad (3)$$

The dependent variable ΔB is the change, between 2007Q2 and 2007Q4, in the fraction of the fund's portfolio represented by securitized bonds (columns 1–2) or corporate bonds (columns 3–4), in excess of the fund sector average. *Turnover ratio* (odd-numbered columns) and *Flow volatility* (even-numbered columns) are as defined in Section 3.2. x is a set of standard mutual fund characteristics (see Appendix A for variable definitions). All explanatory variables are expressed in their values as of June 2007. The standard errors are italicized and appear below coefficients.

Panel B reports the estimates of a model:

$$\Delta C_i = \alpha + \beta \text{ Flow volatility(or turnover)}_i + \gamma' x_i + \varepsilon_i.$$

In the odd-numbered columns, the dependent variable ΔC is defined analogously to ΔB for investment-grade bonds (*High*). In the even-numbered columns, the dependent variable is analogously defined for junk bonds (*Low*). *Turnover ratio* (odd-numbered columns) and *Flow volatility* (even-numbered columns) are as defined in Section 3.2. x is a set of standard control variables as in Panel A. *F*-test statistics and *p*-values for the difference between the *Flow volatility* or *Turnover* coefficients for investment-grade and junk bonds are provided in the row labeled "*F*-stat (*p*-value)." The sample includes all the mutual funds belonging to the merged Lipper eMAXX-CDA/Spectrum data set, over the period 2007Q2–2007Q4. The standard errors are italicized and appear below coefficients. The symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Changes in holdings of corporate bonds and securitized bonds

	Securitized bonds		Corporate bonds	
	(1)	(2)	(3)	(4)
Turnover ratio	−3.4498***		−2.8097***	
	−4.25		*−5.95*	
Flow volatility		−0.1144		−0.3682***
		−0.95		*−3.37*
Secur. holdings 2007Q2	−0.3858***	−0.3969***		
	−10.4	*−10.86*		
Corp. holdings 2007Q2			−0.2664***	−0.2736***
			−10.23	*−10.46*
Affiliated to commercial bank	0.0028	−0.0022	−0.0189	−0.0217*
	0.17	*−0.14*	*−1.63*	*−1.78*
Log(family size)	−0.0017	−0.0029	−0.0006	−0.0003
	−0.51	*−0.9*	*−0.24*	*−0.12*
Past flow	0.2640*	0.3139*	0.2693*	0.5939***
	1.74	*1.7*	*1.89*	*3.62*
Fund return	0.4722	0.4932	0.3035	0.1715
	1.36	*1.45*	*1.32*	*0.71*
Family equity holdings	0.0103	0.0165	−0.0153	−0.0104
	0.4	*0.66*	*−0.7*	*−0.47*
Mgmt fee	−0.0515	−0.0673*	0.0339	0.0236
	−1.4	*−1.87*	*0.81*	*0.54*
Exp. ratio	1.312	2.6289	−6.3101*	−1.8736
	0.39	*0.88*	*−1.79*	*−0.44*
Actual 12b1	−0.8002	−1.5201	6.5211	1.7917
	−0.18	*−0.4*	*1.53*	*0.34*
Av. maturity of holdings	0.0007	0.0004	−0.0010***	−0.0012***
	1.6	*0.93*	*−2.78*	*−3.13*
No equity	0.0183	0.0097	0.0092	0.0002
	0.87	*0.46*	*0.53*	*0.01*
Fund's equity return	−0.0602	−0.0531	−0.0368	0.0055
	−0.31	*−0.28*	*−0.22*	*0.03*
Standard error	White	White	White	White
No. obs.	561	578	561	578
R^2	0.38	0.35	0.3	0.28

Panel B: Sales of corporate bonds, by ratings

Rating	Low	High	Low	High
	(1)	(2)	(3)	(4)
Turnover ratio	−1.4581***	−1.2381***		
	−3.78	*−5.59*		
Flow volatility			−0.2789***	−0.0838*
			−3.34	*−1.92*
[Control variables suppressed]				
F-stat (*p*-value)	0.25 (0.6193)		5.94** (0.0148)	
Standard error	White	White	White	White
No. obs.	561	561	578	578
R^2	0.33	0.1	0.33	0.06

Table 4. Changes in holdings of securitized and corporate bonds.

Interpretation. Table 4 moves from pre-crisis holdings to crisis-period portfolio changes. High-turnover funds reduce both securitized and corporate bond weights, but the flow-volatility measure is especially informative for corporate bond liquidation. In Panel B, the effect of flow volatility is stronger for low-rated corporate bonds than for high-rated corporate bonds, with a significant F-test. This is central for the paper's story because it explains why the shock migrated most strongly into the lower-rated corporate bond market. The funds most exposed to liquidity pressure liquidated the types of corporate bonds that either raised liquidity or reduced risk relative to their now-toxic securitized positions.

5.6. Table 5. Changes in corporate bonds' yield spreads after the onset of the crisis

Table 5
Changes in corporate bonds' yield spreads after the onset of the crisis.

Panel A of this table reports the estimates of a model:

$$\Delta YS_{ij} = \alpha + \beta HoldersExposure_{ij} + \gamma InvRating_{ij} + \delta(HoldersExposure_{ij} \times InvRating_{ij}) + \varphi' x_{ij} + \mu_j + \varepsilon_{ij}, \quad (4)$$

where each observation is a corporate bond with data in the FINRA TRACE Corporate Bond Data. The dependent variable ΔYS is the change in bond i's yield spread, defined as the difference between bond i's yield on the secondary market, as reported by TRACE, and the yield on a Treasury bond of comparable maturity. Data on Treasury bond yields are from the Federal Reserve Statistical Release. In columns 1–4, ΔYS is defined as the change in the bond's yield spread over the period from June to October 2007, while in columns 5–8 it is the change in the bond's yield spread over the period from June to December 2007. The explanatory variables are: *HoldersExposure* (the weighted-average fraction of securitized bonds in the portfolio of the mutual funds that hold bond i, averaged across each investor holding bond i and using par amount of bond i held as weights), *InvRating* (the natural logarithm of the inverse of 1 + the numerical value of the bond's S&P rating, which ranges from zero (no rating) to 24 (AAA rating or higher)), the interaction term between these two variables, and a standard set of bond characteristics x (see Appendix A for variable definitions), issuer fixed effects for issuer j (μ_j), and offering year fixed effects.

Panel B of this table reports the estimates of a model:

$$\Delta YS_{ij} = \alpha + \beta HighExposed_{ij} + \gamma InvRating_{ij} + \delta(HighExposed_{ij} \times InvRating_{ij}) + \varphi' x_{ij} + \mu_j + \varepsilon_{ij}, \quad (5)$$

where each observation is a corporate bond i. The dependent variable ΔYS is defined as in Panel A. *HighExposed* is an indicator equal to one if bond i's high-liquidity-need mutual fund holders' exposure to securitized bonds is above the sample median. We use two proxies for mutual funds' liquidity needs as before (turnover ratio and flow volatility) and obtain one indicator *HighExposed* for each proxy. *InvRating* is defined as in Panel A. We further include the interaction term between these two variables, along with a standard set of bond characteristics x (see Appendix A for variable definitions), issuer fixed effects for issuer j (μ_j), and offering year fixed effects. For brevity, only the key coefficients are shown. In both panels and in all specifications, the standard errors are shown italicized below coefficients and clustered around bond issuers. The symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: The baseline model

	Change in yield spread—July–October 2007				Change in yield spread—July–December 2007			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Holders' exposure (to securitized bonds)	0.7885**	0.6659*	0.5762	0.3936	1.2900*	1.4210**	1.4042*	1.2581*
	2.02	*1.80*	*1.45*	*1.01*	*1.87*	*2.04*	*1.92*	*1.74*
InvRating	0.8066*	0.9079**	0.8497**	0.8048**	1.8664**	1.6608**	1.6482**	1.6086**
	1.84	*2.47*	*2.28*	*2.14*	*2.96*	*2.61*	*2.54*	*2.47*
Holders' exposure × InvRating	0.2911**	0.2576**	0.2988**	0.2568**	0.4836**	0.5349**	0.5438**	0.5311**
	2.48	*2.16*	*2.53*	*2.18*	*2.26*	*2.43*	*2.51*	*2.43*
Yield spread in 2007Q2	−0.1909**	−0.2689***	−0.2717***	−0.2731***	−0.4174***	−0.4134***	−0.4147***	−0.4157***
	−2.48	*−2.89*	*−2.98*	*−2.99*	*−6.70*	*−6.40*	*−6.34*	*−6.37*
No rating [1 = NO]	−2.6943*	−3.0231***	−2.8370**	−2.6829**	−6.0739***	−5.4002***	−5.3617***	−5.2271**
	−1.92	*−2.60*	*−2.41*	*−2.24*	*−3.05*	*−2.70*	*−2.63*	*−2.55*
Bond is not held by mutual funds [1 = NO]	0.0720	0.1387*	0.1188	0.1086	−0.1198	−0.1244	−0.1311	−0.1434
	0.79	*1.95*	*1.57*	*1.56*	*−0.95*	*−0.99*	*−1.11*	*−1.16*
Bond face value	−0.0012	−0.0082	0.0028	0.0120	−0.0372	−0.0333	−0.0302	−0.0225
	−0.07	*−0.60*	*0.19*	*0.79*	*−1.53*	*−1.44*	*−1.01*	*−0.95*
Covenants [1 = YES]	0.1651	0.0267	0.0311	0.0358	−0.1065	−0.0979	−0.0958	−0.0900
	1.37	*0.34*	*0.39*	*0.46*	*−0.73*	*−0.70*	*−0.68*	*−0.65*
CovIndex	−0.2727	−0.1518	−0.1218	−0.1090	0.1924	0.1865	0.1925	0.2066
	−0.87	*−0.73*	*−0.59*	*−0.52*	*0.49*	*0.48*	*0.50*	*0.53*
Log(Months to maturity)	−0.1016***	−0.1039***	−0.1154***	−0.1048***	−0.2147***	−0.2215***	−0.2251***	−0.2220***
	−3.20	*−2.94*	*−3.09*	*−2.85*	*−3.17*	*−3.25*	*−3.14*	*−3.22*
Insurance holders' exposure		0.2657	0.2632	0.2573		−0.2643	−0.2645	−0.2641
		0.75	*0.74*	*0.72*		*−0.52*	*−0.51*	*−0.52*
Insurance holders' exposure × InvRating		0.0374	0.0286	0.0266		−0.0635	−0.0628	−0.0676
		0.33	*0.25*	*0.23*		*−0.37*	*−0.37*	*−0.40*
Amihud			0.1525				0.0468	
			0.94				*0.14*	
Holders' exposure × Amihud			0.6232				0.1272	
			1.35				*0.21*	
InvTrades				0.1873***				0.1088
				2.91				*0.98*
Holders' exposure × InvTrades				−0.2203				−0.1274
				−1.22				*−0.44*
Issuer fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Offering year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 5, first part: baseline yield-spread regressions.

Table 5 (*continued*)

Panel A: The baseline model

	Change in yield spread—July–October 2007				Change in yield spread—July–December 2007			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Standard error cluster	Issuer	Issuer	Issuer	Issuer	Issuer	Issuer	Issuer	Issuer
N. obs.	7,348	7,348	7,348	7,348	8,148	8,148	8,148	8,148
R^2	0.65	0.70	0.70	0.70	0.74	0.75	0.75	0.75

Panel B: High-liquidity-need funds with exposure to securitized bonds

	Change in yield spread—July–October 2007		Change in yield spread—July–December 2007	
	(1)	(2)	(3)	(4)
HighExposed (turnover)	0.3324*		0.7222***	
	1.71		*2.71*	
HighExposed (flow volatility)		0.2429		0.4700**
		1.26		*1.99*
InvRating	0.9439***	0.9231***	1.8800***	1.8093***
	2.79	*2.74*	*3.67*	*3.55*
HighExposed (turnover) × InvRating	0.1181*		0.2083**	
	1.87		*2.54*	
HighExposed (flow vol.) × InvRating		0.1228**		0.1476**
		2.01		*2.09*
Issuer fixed effects	Yes	Yes	Yes	Yes
Other control variables	Yes	Yes	Yes	Yes
Offering year fixed effects	Yes	Yes	Yes	Yes
Standard error cluster	Issuer	Issuer	Issuer	Issuer
No. obs.	7,302	7,302	8,136	8,136
R^2	0.66	0.66	0.76	0.76

Table 5, continued: high-liquidity-need exposed funds.

Interpretation. Table 5 is the core asset-pricing result. Corporate bonds held by mutual funds with greater securitized-bond exposure experience larger yield-spread increases during the crisis. The positive interaction between holders' exposure and InvRating means the effect is stronger for lower-rated corporate bonds. Importantly, the regressions include issuer fixed effects, so the result compares bonds issued by the same firm but held by investors with different securitized-bond exposure. This sharply reduces the concern that exposed investors merely held worse firms. Insurance-company exposure is generally not significant, which points toward the mutual-fund liquidity channel rather than a broad institutional exposure channel. Panel B confirms that the effect is especially visible when exposed holders are high-liquidity-need mutual funds. **Method note.** The paper interprets the estimates as economically large: moving from 0% to 50% mutual-fund exposure to securitized bonds is associated with a substantially larger rise in corporate-bond yield spreads, especially for lower-rated bonds.

5.7. Table 6. Corporate bond selling pressure by mutual funds

Table 6
Corporate bond selling pressure by mutual funds.

Panel A reports the estimates of a model:

$$\Delta Tr_{ij} = \alpha + \beta HoldersExposure_{ij} + \gamma InvRating_{ij} + \delta(HoldersExposure_{ij} \times InvRating_{ij}) + \varphi' x_{ij} + \mu_j + \varepsilon_{ij}, \qquad (6)$$

where each observation is a corporate bond with data in the FINRA TRACE Corporate Bond Data. The dependent variable ΔTr_i is defined as the total net sales of bond i by mutual funds divided by the total trading volume on the bond, over the crisis period. In columns 1–4, ΔTr_i is defined over the period from July to October 2007, while in columns 5–8 it is defined over the period from July to December 2007. The explanatory variables are: *HoldersExposure* (the weighted-average fraction of securitized bonds in the portfolio of the mutual funds that hold bond i, averaged across each investor holding bond i and using par amount of bond i held as weights), *InvRating* (the natural logarithm of the inverse of 1 + the numerical value of the bond's S&P rating, which ranges from zero (no rating) to 24 (AAA rating or higher)), the interaction term between these two variables, and a standard set of bond characteristics x (see Appendix A for variable definitions), issuer fixed effects for issuer j (μ_j), and offering year fixed effects.

Panel B reports the estimates of a model:

$$\Delta Tr_{ij} = \alpha + \beta HighExposed_{ij} + \gamma InvRating_{ij} + \delta(HighExposed_{ij} \times InvRating_{ij}) + \varphi' x_{ij} + \mu_j + \varepsilon_{ij}, \qquad (7)$$

where each observation is a corporate bond. The dependent variable ΔTr is defined as in Panel A. *HighExposed* is an indicator equal to one if the bond is held by high-liquidity-need mutual funds with exposure to securitized bonds above the median. We use two proxies for mutual funds' liquidity needs (turnover ratio and flow volatility) and obtain one indicator *HighExposed* for each proxy. *InvRating* is defined as in Panel A. We further include the interaction term between these two variables, along with a standard set of bond characteristics x (see Appendix A for variable definitions), issuer fixed effects for issuer j (μ_j), and offering year fixed effects. For brevity, only the key coefficients are shown. In both panels and in all specifications, the standard errors are shown italicized below coefficients and clustered around bond issuers. The symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Baseline model

	Trading pressure of mutual funds—July–October 2007				Trading pressure of mutual funds—July–December 2007			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Holders' exposure	0.0056	0.0199	0.0198	0.0176	0.0312	0.0412	0.0506	0.0552
	0.21	*0.65*	*0.68*	*0.52*	*0.99*	*1.20*	*1.55*	*1.46*
InvRating	0.0151	0.0148	0.0163	0.0176	0.0102	0.0105	0.0127	0.0093
	0.78	*0.77*	*0.84*	*0.93*	*0.55*	*0.57*	*0.69*	*0.51*
Holders' exposure × InvRating	0.0105	0.0154	0.0065	0.0118	0.0238**	0.0273**	0.0189*	0.0251**
	1.10	*1.39*	*0.65*	*1.25*	*2.37*	*2.44*	*1.82*	*2.49*
2007Q2 Log-volume	0.0022**	0.0024**	0.0015	0.0006	0.0014**	0.0013**	0.0012**	0.0014*
	2.03	*2.16*	*1.11*	*0.54*	*2.13*	*2.08*	*2.36*	*1.90*
No rating [1=NO]	−0.0518	−0.0495	−0.0552	−0.0601	−0.0352	−0.0351	−0.0430	−0.0325
	−0.84	*−0.81*	*−0.89*	*−0.98*	*−0.59*	*−0.60*	*−0.73*	*−0.55*
Bond is not held by mutual funds [1=NO]	−0.0461***	−0.0465***	−0.0469***	−0.0462***	−0.0562***	−0.0559***	−0.0555***	−0.0566***
	−9.25	*−9.27*	*−9.43*	*−9.22*	*−12.01*	*−12.07*	*−11.81*	*−11.88*
Bond face value	0.0006	0.0006	0.0005	0.0002	0.0015**	0.0014**	0.0010*	0.0016**
	0.86	*1.00*	*0.74*	*0.23*	*2.44*	*2.22*	*1.66*	*2.39*
Covenants [1=YES]	0.0104	0.0115	0.0109	0.0095	0.0079	0.0086	0.0073	0.0086
	1.14	*1.26*	*1.21*	*1.06*	*0.80*	*0.88*	*0.76*	*0.87*
CovIndex	−0.0012	−0.0056	−0.004	−0.0019	0.0140	0.0104	0.0107	0.0150
	−0.05	*−0.24*	*−0.17*	*−0.08*	*0.54*	*0.41*	*0.42*	*0.58*
Log(months to maturity)	−0.0018**	−0.0018**	−0.0015*	−0.0018**	−0.0052***	−0.0052***	−0.0044***	−0.0052***
	−2.21	*−2.21*	*−1.67*	*−2.23*	*−5.09*	*−5.11*	*−4.12*	*−5.09*
Insurance holders' exposure		−0.022				−0.0090		
		−1.44				*−0.50*		
Insurance holders' exposure × InvRating		−0.0064				−0.0048		
		−1.19				*−0.75*		
Amihud			−0.0013				−0.0059*	
			−0.28				*−1.85*	
Holders' exposure × Amihud			−0.0738**				−0.0946**	
			−1.98				*−2.51*	
InvTrades				−0.0106***				0.0001
				2.83				*0.03*
Holders' exposure × InvTrades				0.0062				−0.0205
				0.39				*−1.35*
Issuer fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Offering year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Standard error cluster	Issuer	Issuer	Issuer	Issuer	Issuer	Issuer	Issuer	Issuer

Table 6, first part: baseline mutual-fund selling pressure.

Table 6 (*continued*)

Panel A: Baseline model

	Trading pressure of mutual funds—July–October 2007				Trading pressure of mutual funds—July–December 2007			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
No. obs.	9,042	9,042	9,005	9,042	8,728	8,728	8,728	8,728
R^2	0.69	0.69	0.69	0.69	0.68	0.68	0.68	0.68

Panel B: High-liquidity-need funds with exposure to securitized bonds

	Mutual fund selling pressure—July–October 2007		Mutual fund selling pressure—July–December 2007	
	(1)	(2)	(3)	(4)
HighExposed (turnover)	0.0383***		0.0490***	
	2.86		*4.02*	
HighExposed (flow volatility)		0.0375***		0.0436***
		2.82		*2.98*
InvRating	0.0200	0.0188	0.0197	0.0183
	0.77	*0.73*	*1.08*	*1.01*
HighExposed (turnover) × InvRating	0.0096**		0.0073*	
	2.14		*1.84*	
HighExposed (flow vol.) × InvRating		0.0079*		0.0067
		1.86		*1.52*
Issuer fixed effects	Yes	Yes	Yes	Yes
Other control variables	Yes	Yes	Yes	Yes
Offering year fixed effects	Yes	Yes	Yes	Yes
Standard error cluster	Issuer	Issuer	Issuer	Issuer
No. obs.	8,666	8,666	8,728	8,728
R^2	0.58	0.58	0.69	0.68

Table 6, continued: high-liquidity-need exposed funds.

Interpretation. Table 6 connects the price effect in Table 5 to actual trading pressure. The dependent variable measures mutual-fund net selling pressure relative to trading volume. Exposed mutual fund holders, especially high-turnover and high-flow-volatility holders, are associated with greater selling pressure. This is an important mechanism check: the spread results do not simply reflect investors marking down bonds they held; the same exposed investor base is linked to observable net selling in the secondary market. The interaction terms again indicate that lower-rated bonds were more affected.

5.8. Table 7. Corporate bond trading volume after the onset of the crisis

Table 7
Corporate bond trading volume after the onset of the crisis.
The table reports the estimates of a model:

$$LogVol_{ij} = \alpha + \beta HoldersExposure_{ij} + \gamma InvRating_{ij} + \delta(HoldersExposure_{ij} \times InvRating_{ij}) + \varphi' x_{ij} + \mu_j + \varepsilon_{ij},$$

where each observation is a corporate bond with data in the FINRA TRACE Corporate Bond Data. The dependent variable *LogVol* is the natural logarithm of bond *i*'s average daily trading volume (expressed in number of trades) over the crisis period. In columns 1–4, *LogVol* is defined over the period from July to October 2007, while in columns 5–8 it is defined over the period from July to December 2007. The explanatory variables are: *HoldersExposure* (the weighted-average fraction of securitized bonds in the portfolio of the mutual funds that hold the bond *i*, averaged across each investor holding bond *i* and using par amount of bond *i* held as weights), *InvRating* (the natural logarithm of the inverse of 1 + the numerical value of the bond's S&P rating, which ranges from zero (no rating) to 24 (AAA rating or higher)), the interaction term between these two variables, and a standard set of bond characteristics x (see Appendix A for variable definitions), issuer fixed effects for issuer $j(\mu_j)$, and offering year fixed effects. In all specifications, the standard errors are shown italicized below coefficients and clustered around bond issuers. The symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Trading volume—July–October 2007				Trading volume—July–December 2007			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Holders' exposure (to securitized bonds)	0.8778***	0.9275***	1.3250***	0.5752	0.6254***	0.6348**	1.0685***	−0.0996
	2.73	*2.63*	*3.49*	*1.34*	*2.96*	*2.49*	*3.75*	*−0.33*
InvRating	−1.5356***	−1.5018***	−1.4736***	−1.4969***	−0.6168***	−0.5842***	−0.5456***	−0.5862***
	−4.38	*−4.37*	*−4.35*	*−4.36*	*−2.96*	*−2.88*	*−2.78*	*−2.92*
Holders' exposure × InvRating	0.2382**	0.2686**	0.1395	0.2664**	0.1564**	0.1718*	0.0253	0.1594*
	2.28	*2.21*	*1.11*	*2.21*	*2.18*	*1.89*	*0.30*	*1.76*
2007Q2 Log − volume	0.4286***	0.4152***	0.4076***	0.4176***	0.5093***	0.4963***	0.4588***	0.5088***
	10.44	*10.50*	*11.02*	*11.08*	*18.22*	*18.03*	*17.28*	*19.98*
No rating [1 = NO]	4.8706***	4.7862***	4.7075***	4.7739***	2.0144***	1.9284***	1.8119***	1.9423***
	4.46	*4.46*	*4.44*	*4.45*	*3.11*	*3.05*	*2.95*	*3.10*
Bond is not held by mutual funds [1 = NO]	−0.6231***	−0.5944***	−0.6176***	−0.5910***	−0.4955***	−0.4686***	−0.4976***	−0.4627***
	−6.10	*−5.89*	*−6.03*	*−5.83*	*−6.63*	*−6.37*	*−6.65*	*−6.37*
Bond face value	0.2756***	0.2649***	0.2636***	0.2653***	0.1389***	0.1289***	0.1260***	0.1317***
	7.62	*7.37*	*7.25*	*7.07*	*8.74*	*8.28*	*8.12*	*8.07*
Covenants [1 = YES]	0.0862	0.1031	0.1084	0.0931	0.1788*	0.1914**	0.1942**	0.1773*
	0.64	*0.78*	*0.83*	*0.70*	*1.90*	*2.08*	*2.18*	*1.94*
CovIndex	0.6117**	0.5301*	0.4661	0.5293*	0.4216**	0.3590*	0.2972	0.3563*
	2.07	*1.81*	*1.60*	*1.81*	*2.27*	*1.94*	*1.64*	*1.93*
Log(Months to maturity)	−0.0981***	−0.0958***	−0.0928***	−0.0954***	−0.1056***	−0.1035***	−0.0869***	−0.1028***
	−4.57	*−4.42*	*−3.88*	*−4.44*	*−8.47*	*−8.32*	*−5.97*	*−8.47*
Insurance holders' exposure		0.3089	0.3357	0.3042		0.3452*	0.3919**	0.3325*
		0.97	*1.07*	*0.95*		*1.77*	*2.09*	*1.65*
Insurance holders' exposure × InvRating		−0.0785	−0.0612	−0.0796		−0.0552	−0.0302	−0.0589
		−0.69	*−0.54*	*−0.70*		*−0.88*	*−0.50*	*−0.93*
Amihud			0.0632				−0.1411	
			0.33				*−1.50*	
Holders' exposure × Amihud			−2.2277**				−2.4864***	
			−2.46				*−4.18*	
InvTrades				−0.0400				−0.1293**
				−0.50				*−2.50*
Holders' exposure × InvTrades				0.3435				0.6824***
				1.41				*4.17*
Issuer fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Offering year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Standard error cluster	Issuer	Issuer	Issuer	Issuer	Issuer	Issuer	Issuer	Issuer
No. obs.	9,598	9,598	9,598	9,598	9,573	9,573	9,573	9,573
R^2	0.80	0.80	0.80	0.80	0.88	0.88	0.88	0.88

Table 7. Corporate bond trading volume after the onset of the crisis.

Interpretation. Table 7 shows that exposed bonds also experienced abnormal trading volume. Holders' exposure is positively related to trading volume, especially in the early crisis window and in specifications with lower-rated-bond interactions. This provides another layer of evidence for fire-sale style pressure. If the result were only a valuation effect, we would not necessarily expect trading volume to move in the same direction. Higher volume, together with Table 6's selling-pressure evidence and Table 5's yield-spread increases, supports a coherent chain: exposed holders faced liquidity stress, sold corporate bonds, and the affected bonds traded more and repriced downward.

5.9. Figure 2. Corporate bond yield spreads and LIBOR-OIS spread

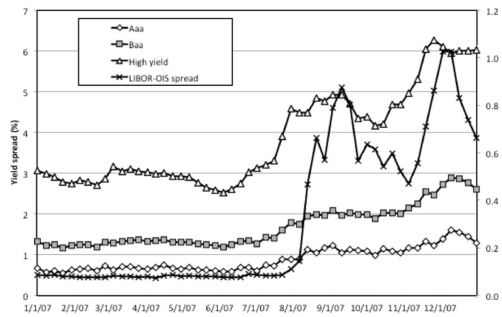

Fig. 2. Corporate bond yield spreads and LIBOR-OIS spread. This figure plots weekly average yield spreads on corporate bonds in 2007, for Aaa, Baa, and high yield corporate bonds, as well as the spread between the LIBOR rate and the Overnight Index swap rate (LIBOR-OIS Spread, secondary axis). The yield spread is defined as the difference between a bond's yield on the secondary market and the yield on a Treasury bond of comparable maturity. Data on Treasury yields are retrieved from the Federal Reserve Statistical Release; corporate bond index yield data used are the Barclays US Aggregate Indexes and are retrieved from Datastream; and the LIBOR-OIS data are retrieved from Bloomberg.

Figure 2. Corporate bond yield spreads and LIBOR-OIS spread.

Interpretation. Figure 2 provides the macro-financial background. The LIBOR-OIS spread jumps sharply around August 2007, which marks the funding-liquidity shock. At the same time, corporate bond spreads widen, with the high-yield series moving much more than Aaa or Baa spreads. This motivates the paper's focus on lower-rated corporate bonds: the crisis did not affect all corporate bonds equally, and the strongest dislocation appeared in the segment most consistent with mutual-fund selling pressure and risk reduction.

5.10. Figure 3. Cumulative return of a long-short portfolio of low-rated corporate bonds with and without exposure

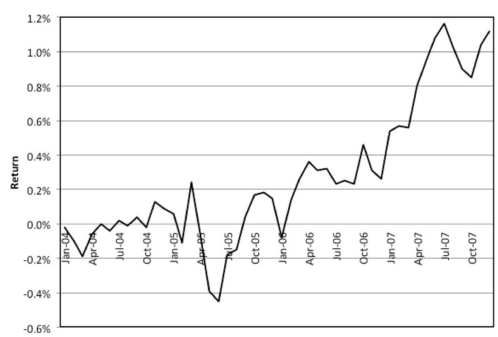

Fig. 3. Cumulative return of a long-short portfolio of low-rated corporate bonds with and without exposure. This figure plots the cumulative monthly return on a portfolio that is *short* on the below-investment-grade bonds whose mutual fund holders have high exposure to securitized bonds, and *long* on a set of issuer- and duration-matched bonds without the exposure, over the period 2004–2007. We define "high exposure" as those corporate bonds whose average mutual fund holder's exposure to securitized bonds is in the top 30% in either of the previous two quarters. We place these bonds in the short portfolio if and only if it has a matching bond without a high exposure satisfying the following criteria: (i) the matching bond is issued by the same issuer firm; and (ii) the time to maturity of the matching bond is between 50% and 150% of the time to maturity of the shorted bond. We place these matching bonds in the long portfolio. The long-short portfolio's monthly return is then the long portfolio's monthly return (rebalanced to be equal-weighted each month) minus the short portfolio's monthly return (similarly rebalanced). Returns on individual corporate bonds are constructed from the secondary market prices, as reported by TRACE. In each period, bond returns are also winsorized at 1%.

right-hand side panel, which is consistent with the We also augment specification (4) by adding the

Figure 3. Cumulative return of a long-short portfolio of low-rated corporate bonds with and without exposure.

Interpretation. Figure 3 is an intuitive portfolio-based demonstration of the main result. The portfolio is short low-rated corporate bonds whose mutual-fund holders had high exposure to securitized bonds and long issuer- and duration-matched bonds without that exposure. The cumulative return rises sharply during 2007, meaning the exposed low-rated bonds underperform their matched unexposed counterparts. This figure is useful because it visualizes the same-issuer identification: the comparison is not simply between risky firms and safe firms, but between similar bonds whose difference is the pre-crisis portfolio exposure of their holders.

5.11. Table 8. Relation between mutual funds' and insurance companies' trades

Table 8

Relation between mutual funds' and insurance companies' trades.

Panel A reports the estimates of a model:

$$MF\_Netbuy_{it} = \alpha + \beta_1 INS\_Netbuy_{it} \times (1 - Crisis_t)\beta_2 INS\_Netbuy_{it} \times Crisis_t + \gamma' x_{it-1} + \varepsilon_{it}, \quad (8)$$

where each observation is a corporate bond i in quarter t. The dependent variable *MF_Netbuy* is the net purchases of the bond by all mutual funds in column 1. In columns 2 and 3 they are the net purchases by high-liquidity-needs, exposed mutual funds (i.e., funds that have high liquidity needs with exposure to securitized bonds above the median). We use two proxies for high liquidity needs: High turnover (column 2) and High flow volatility (column 3). In columns 4 and 5 they are the net purchases by all funds except the high-liquidity-needs, exposed funds. *MF_Netbuy* is calculated as the net purchases of bond i by the funds divided by the total institutional holdings of the bond (holdings of mutual funds plus holdings of insurance companies) as of the previous quarter. *INS_Netbuy* is the net purchases of the bond by insurance companies, again divided by the total institutional holdings of the bond. *Crisis* is an indicator equal to one for dates between 2007Q3 and 2008Q1. x is a vector of standard bond characteristics (see Appendix A for variable definitions). The last row of the table reports the F-test statistic for H_0: $\beta_1 = \beta_2$. The sample is for the period 1998Q1–2008Q1.

Panel B reports the estimates of a model:

$$MF\_Netbuy_i = \alpha + \beta\ INS\_Netbuy_i + \gamma' x_i + \varepsilon_i,$$

where *MF_Netbuy* is the net purchases of bond i by funds that experience below-median flows over the last six months of 2007 (*LowFlow* funds) in column 1 and *HighFlow* funds in column 2. *INS_Netbuy* is defined as above, and x is a set of standard bond characteristics (see Appendix A for variable definitions), including offering year effects. The standard errors are italicized and appear below coefficients. In both panels, the symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Correlations of mutual fund and insurance company trades, before and after the onset of the crisis

	All funds	High turnover	High flow volatility	Non-high turnover	Non-high volatility
	(1)	(2)	(3)	(4)	(5)
INS_Netbuy × (1 − Crisis)	0.0892***	0.0010***	0.0016***	0.0862***	0.0835***
	11.18	*5.05*	*5.03*	*10.75*	*10.56*
INS_Netbuy × Crisis	0.2075***	0.0041***	0.0068***	0.1789***	0.1735***
	8.86	*5.03*	*5.29*	*7.69*	*7.81*
[Control variables suppressed]					
Bond and quarter fixed effects	Y	Y	Y	Y	Y
Standard error cluster	Bond	Bond	Bond	Bond	Bond
No. obs.	68,233	67,744	67,718	67,077	67,069
R^2	0.06	0.05	0.05	0.06	0.06
F-stat for H_0: $\beta_1=\beta_2$	23.00***	15.51***	14.01***	14.69***	14.39***

Panel B: Trade correlations and flows after the onset of the crisis

	LowFlow funds		HighFlow funds
	(1)		(2)
INS_Netbuy	0.0008***		0.0017***
	2.76		*2.67*
[Control variables suppressed]			
Offering year fixed effect	Y		Y
Standard error	White		White
No. obs.	9,598		9,539
R^2	0.02		0.02
F-stat for H_0: $\beta_{LowFlow}=\beta_{HighFlow}$		5.75 (0.0165)	

Table 8. Relation between mutual funds' and insurance companies' trades.

Interpretation. Table 8 tests whether insurance companies acted as strategic liquidity providers who offset mutual-fund sales. The positive relationship between insurance-company net purchases and mutual-fund net purchases becomes stronger during the crisis rather than reversing. If insurers were fully absorbing mutual-fund fire sales, one would expect a negative relation between mutual-fund and insurance-company trades. Instead, the evidence suggests that insurers did not provide enough countervailing liquidity. Panel B also shows that the relationship differs across low-flow and high-flow mutual funds, reinforcing that flow pressure matters for trading behavior.

5.12. Table 9. The structural break in institutional trades and the correlation of the yield spread to trades

Table 9

The structural break in institutional trades and the correlation of the yield spread to trades.

In columns 1-2, the table reports the estimates of a model:

$$Netbuy_{it} = \beta_1(1-Crisis_{it}) + \beta_2\ Crisis_{it} + \varepsilon_{it},$$

where *Netbuy* is either *INS_Netbuy* (column 1) or *MF_Netbuy* (column 2). *INS_Netbuy* is the aggregate net purchases of bond *i* by all insurance companies, divided by the prior-quarter total holdings of insurance companies plus mutual funds. *MF_Netbuy* is analogously defined. *Crisis* is an indicator variable equal to one for dates between 2007Q3 and 2008Q1. The last row reports the *F*-test statistic for H_0: $\beta_1 = \beta_2$.

In column 3, the table reports the estimates of a model:

$$\Delta YS_{it} = \alpha + \beta_1\ INST\_Netbuy_{it}(1-Crisis_t) + \beta_2\ INST\_Netbuy_{it} \times Crisis_t + \gamma' x_{it} + \varepsilon_{it},$$

where *INST_Netbuy* is the sum of *INS_Netbuy and MF_Netbuy*, *Crisis* is as defined above, and *x* is a set of standard bond characteristics (see Appendix A for variable definitions), including bond and quarter fixed effects. The last row reports the *F*-test statistic for H_0: $\beta_1 = \beta_2$.

In columns 4–5, the table reports the estimates of a model:

$$\Delta YS_{it} = \alpha + \beta_1\ INS\_Netbuy_{it}(1-Crisis_t) + \beta_2\ INS\_Netbuy_{it} \times Crisis_t + \beta_3\ MF_{it}(1-Crisis_t) + \beta_4\ MF_{it} \times Crisis_t + \gamma' x_{it} + \varepsilon_{it},$$

where *Crisis* and *INS_Netbuy* are defined as above, and *MF* is either *MF_Netbuy* (column 4), defined as above, or *LowFlow_Netbuy* (column 5), defined as the net purchases of bond *i* by mutual funds that experience below-median flows in the quarter, divided by the prior-quarter total holdings of insurance companies plus mutual funds. *x* is a set of standard bond characteristics (see Appendix A for variable definitions), including individual bond and quarter fixed effects. The last row reports the *F*-test statistic for H_0: $\beta_3 = \beta_4$. The standard errors are italicized and appear below coefficients. The sample is for the period 1998Q1–2008Q1. The symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable	INS_Netbuy	MF_Netbuy	ΔYS	ΔYS	ΔYS
	(1)	(2)	(3)	(4)	(5)
1 − Crisis	−0.0302***	−0.0233***			
	−109.75	*−66.95*			
Crisis	−0.0266***	−0.0436***			
	−54.88	*−53.83*			
INST_Netbuy(1 − Crisis)			0.1099		
			0.80		
INST_Netbuy × Crisis			−2.3966***		
			−9.86		
INS_Netbuy(1 − Crisis)				−0.1709	−0.1258
				−1.39	*−1.01*
INS_Netbuy × Crisis				−0.6031*	−0.6001*
				−1.83	*−1.78*
MF_Netbuy(1 − Crisis)				0.6210***	
				3.51	
MF_Netbuy × Crisis				−3.4781***	
				−12.11	
LowFlow_Netbuy(1 − Crisis)					−0.3964
					−1.01
LowFlow_Netbuy × Crisis					−5.1208***
					−5.73
[Control variables suppressed]					
Bond and quarter fixed effects	N	N	Y	Y	Y
Standard error cluster	Bond	Bond	Bond	Bond	Bond
No. obs.	63,330	63,757	63,520	63,137	62,231
R^2	0.23	0.19	0.11	0.16	0.15
F-stat (*p*-value)	6.19 (0.01)	719.11 (0.00)	84.19 (0.00)	153.36 (0.00)	22.48 (0.00)

Table 9. Structural break in institutional trades and the correlation of yield spreads to trades.

Interpretation. Table 9 documents a structural break in the market impact of institutional trades. During the crisis, net purchases by institutional investors become much more strongly related to yield-spread changes. The negative crisis coefficients in the spread regressions mean that selling pressure is associated with rising spreads. The mutual-fund coefficients are particularly large, and the LowFlow-fund measure shows that funds with worse flows have especially strong price impact. This table ties the paper's transmission story to market liquidity: trades mattered more during the crisis because the market's capacity to absorb sales deteriorated.

6. How the Results Fit Together

The empirical evidence is cumulative rather than dependent on a single regression. Figure 1 and Table 3 show that mutual funds with high liquidity needs held large securitized-bond positions before the crisis. Table 2 shows that the most flow-stressed funds sold corporate bonds rather than securitized bonds at the onset of the crisis. Tables 4, 6, and 7 connect high-liquidity-need funds to selling pressure and trading activity in corporate bonds. Table 5 and Figure 3 show that those sales had price consequences, especially for lower-rated corporate bonds. Tables 8 and 9 show that insurance companies did not fully offset this pressure and that institutional trades had stronger price impact during the crisis.

7. Interpretation of the Crisis Transmission Channel

The paper's mechanism can be summarized in one sentence: *a shock to the liquidity of securitized bonds forced exposed mutual funds to sell corporate bonds, transmitting the crisis across asset classes.* The key insight is that investors do not always sell the asset hit by the shock. When the shocked asset becomes too illiquid, they may sell other assets. This can make safe-looking portfolio connections dangerous: assets that do not share fundamentals can become linked through common ownership and liquidity management.

8. Strengths, Caveats, and Takeaways

8.1. Strengths

- The data allow direct measurement of institutional portfolios rather than relying only on prices.
- The use of issuer fixed effects is a strong identification feature because it holds constant firm-level fundamentals.
- The paper tests multiple implications of the mechanism: holdings, flows, selling, volume, spread changes, and investor-type differences.
- The findings connect micro-level investor behavior to macro-level crisis contagion.

8.2. Caveats

- The data observe important institutional investors but not every potential holder, such as hedge funds and some foreign investors.
- The paper identifies a transmission channel, not the entire explanation for corporate bond spread movements in 2007.
- Holder exposure may still be correlated with some unobserved bond-specific characteristics, though issuer fixed effects and bond controls reduce this concern substantially.
- The mechanism is strongest for the early crisis period and for lower-rated bonds; it should not be interpreted as implying identical effects across all fixed-income markets.

8.3. What to remember

1. Securitized bonds were not the main assets sold by mutual funds after they became toxic; corporate bonds were.
2. Mutual funds with worse flows and greater liquidity needs sold more corporate bonds.

3. Bonds held by investors with greater securitized-bond exposure suffered larger spread increases, especially if lower rated.
4. Insurance companies did not provide enough offsetting liquidity.
5. Common ownership can transmit shocks across markets even when the assets have different fundamentals.